

Design and Implementation of Audio Amplifier



INSTITUTE OF RADIO PHYSICS & ELECTRONICS
UNIVERSITY OF CALCUTTA

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Ritam Pal

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1. Introduction To Audio Amplifier:

An audio amplifier is an essential electronic device designed to increase the amplitude of low-level audio signals to a level suitable for driving loudspeakers or other output devices. It plays a critical role in audio systems, enabling weak signals from sources such as microphones, musical instruments, or audio playback devices to be converted into audible sound with sufficient volume and clarity.

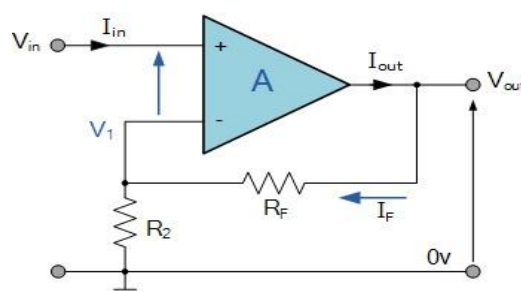
At its core, the audio amplifier operates based on the principles of electronic amplification. It utilizes active components such as transistors or operational amplifiers to modulate and boost the input signal while maintaining its original characteristics such as frequency and waveform shape. The key objective is to amplify the signal without introducing significant distortion or noise.

Audio amplifiers are broadly categorized into different classes—such as Class A, B, AB, and D—each with distinct characteristics in terms of efficiency, linearity, and complexity. The choice of amplifier class and design depends on the intended application, power requirements, and quality considerations. In our project we designed class AB amplifier.

2. Pre Amplifier:

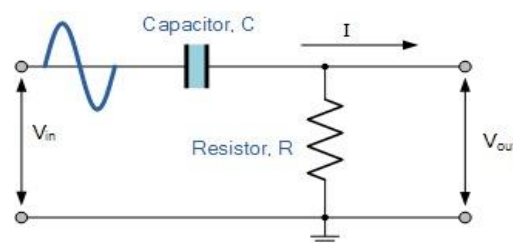
The UA741 operational amplifier is used as a non-inverting voltage amplifier to boost low-level audio signals. In this configuration, the input signal is applied to the non-inverting terminal, ensuring high input impedance and minimal loading on the audio source. The gain is determined by the resistor network as $A_v = 1 + \left(\frac{R_f}{R_2}\right)$;

providing stable amplification with low distortion. This stage prepares the signal to adequately drive the Class AB output stage for higher power amplification.



3. High Pass Filter:

A High Pass Filter (HPF) allows signals with frequencies higher than the cutoff frequency (f_c) to pass while attenuating lower frequencies. It is formed by connecting a capacitor 'C' in series with the input and a resistor 'R' in parallel to the output. The cutoff frequency is given by $f_c = 1/2\pi RC$. At frequencies below f_c , the capacitor blocks low-frequency signals by acting as an open circuit, while at higher frequencies it allows AC signals to pass. HPFs are widely used in audio systems to block DC offsets and low-frequency noise.



4. Class AB Amplifier:

A Class AB amplifier combines the advantages of Class A and Class B configurations to provide higher efficiency with reduced crossover distortion. In a Class B amplifier, two complementary transistors (NPN and PNP) conduct alternately for each half of the input signal, but this leads to distortion near the zero-crossing point due to the base-emitter threshold voltage (V_{BE}) of the transistors. To overcome this, Class AB amplifiers introduce a small bias current so that both transistors conduct slightly even at zero input, eliminating the dead zone.

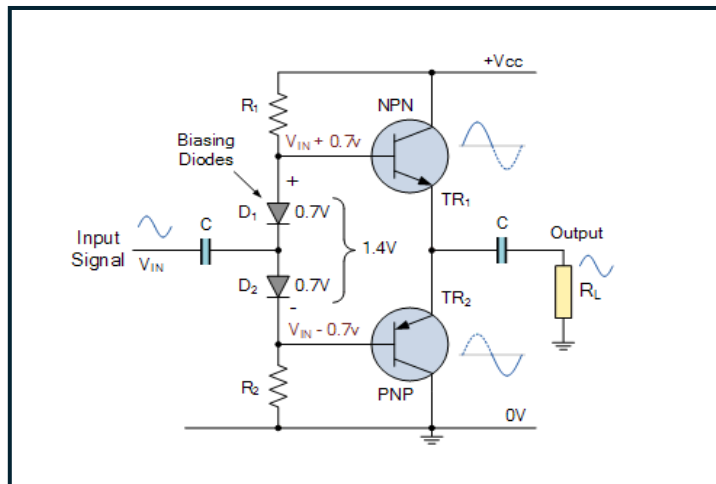
Diode biasing is a simple and effective method to achieve this small quiescent current. Two silicon diodes are connected in series between the bases of the complementary transistors (BD139 and BD140 in this design). Since each diode has a forward voltage drop of approximately 0.7 V, the total bias voltage is about 1.4 V. This voltage compensates for the base-emitter voltage drop of the transistors, keeping them slightly turned on.

The quiescent current through the output stage can be approximated as :

$$I_Q = \frac{V_{bias} - 2V_{BE}}{2R_E}$$

where R_E is the emitter resistor. This biasing minimizes crossover distortion while preventing excessive idle current that could lead to thermal runaway.

Class AB Amplifier



5. Design Process:

In our design we use 5W 4Ω speaker. So, V_{rms} across the speaker should be

$$\begin{aligned} V_{rms} &= \sqrt{P_L} \cdot \sqrt{R_L} \\ &= \sqrt{5 * 4} \quad P_L = 5W \text{ and } R_L = 4\Omega \\ V_{rms} &= 4.47 V \end{aligned}$$

Now,

$$\begin{aligned} V_{peak} &= \sqrt{2} * 4.47 = 6.3245 V \\ I_{peak} &= \frac{6.3245}{4} = 1.58A \end{aligned}$$

We have chosen to design with BD139 and BD140 which have a collector current limit of 1.5A. So, we have added one emitter resistance to limit the current 11 Ω resistances are available with us so the peak current drops to $I_{peak} = \frac{12}{11+4} = 0.8 A$.

Voltage at speaker, $(V_{peak})_{speaker} = I_{peak} * R_L$

$$= 0.8 * 4 = 3.2V$$

$$(V_{rms})_{speaker} = \frac{3.2}{\sqrt{2}} = 2.262 V$$

Power deliver to the speaker,

$$(P)_{speaker} = \frac{(V_{rms})^2}{R_L} = \frac{2.262^2}{4} = 1.279 W \approx 1.3 W$$

Power dissipated in emitter resistor,

IRMS through 11 Ω resistance,

$$I_{RMS} = \frac{I_{peak}}{\sqrt{2}} = \frac{0.8}{\sqrt{2}} = 0.565 A$$

∴ Power dissipation

$$P = I_{rms}^2 * R_E = (0.565)^2 * 11 = 3.52 W$$

Total power drawn = $V_{CC} * I_{peak}$

$$= 12 * 0.8 = 9.6 W$$

Total Power dissipations in transistors = $(9.6 - 3.52 - 1.3) W = 4.78 W$

So each transistors will dissipate around 2.4W of power.

Voltage drop across each transistor's collector-emitter junction is

$$= \frac{P}{I_{rms}} = \frac{2.4}{0.56} = 4.28V$$

Power dissipation in base voltage divider –

Two ($680\ \Omega \parallel 680\ \Omega$) are in series across 12V

$$\therefore \text{current through it, } I = \frac{12}{680}$$

$$I = 17.64\ \text{mA}$$

So power dissipated in both biasing resistors

$$P_{bais} = 2 * (0.0176)^2 * 340 = 0.2107\ \text{W}$$

Negligible power dissipation at pre amplifier section. So, we consider total power dissipation

$$P_{total} = (9.6 + 0.217) = 9.817\ \text{W}$$

$$\therefore \text{Efficiency} = \frac{P_{in}}{P_{out}} * 100\%$$

$$= \frac{P_{speaker}}{P_{total}} * 100\%$$

$$= \frac{1.3}{9.8} * 100\% = 13.3\%$$

We have neglected the power dissipation across the diodes. Before the pre amplifier section. We have considered one HPF to avoid some huge bass coming from input. We want to design a HPF with cut off frequency 150Hz but whatever components are available with us are $R=2000\ \Omega$ and we have used two 1uF capacitors in the series to form $c = 0.5\ \mu\text{F}$. So, we have the cutoff frequency $f_c = 159.15\ \text{HZ} \approx 160\text{HZ}$.

In the pre-amplifier section, we have used a non-inverting type of amplifier so that the gain of amplifier is, $A = 1 + \left(\frac{R1}{R2}\right)$.

$$A = 1 + \left(\frac{82K\Omega}{10K\Omega}\right) = 9.2$$

Generally input audio signal voltage is 0.2V and we have used uA741 which have input impedance $1M\Omega$.

$$\text{So, current } I_{in} = \frac{0.2}{10^6} = 0.2 \mu A$$

$$\text{Power at input} = \frac{V^2}{R} = \frac{0.1^2}{500\Omega} = 80 \mu W$$

We have added one 22 μF capacitor between Pre-Amplifier and class AB Amplifier to block any DC input.

We have driven 0.8A current from the collector and h_{FE} nearly 50 so required base current is $\frac{0.8}{50} = 16 \text{ mA}$. Which is slightly lower than the biasing current.

Now base current becomes $= \frac{5.29+1.84}{340} = 21 \text{ mA}$. Which is enough to drive the class AB's transistor.

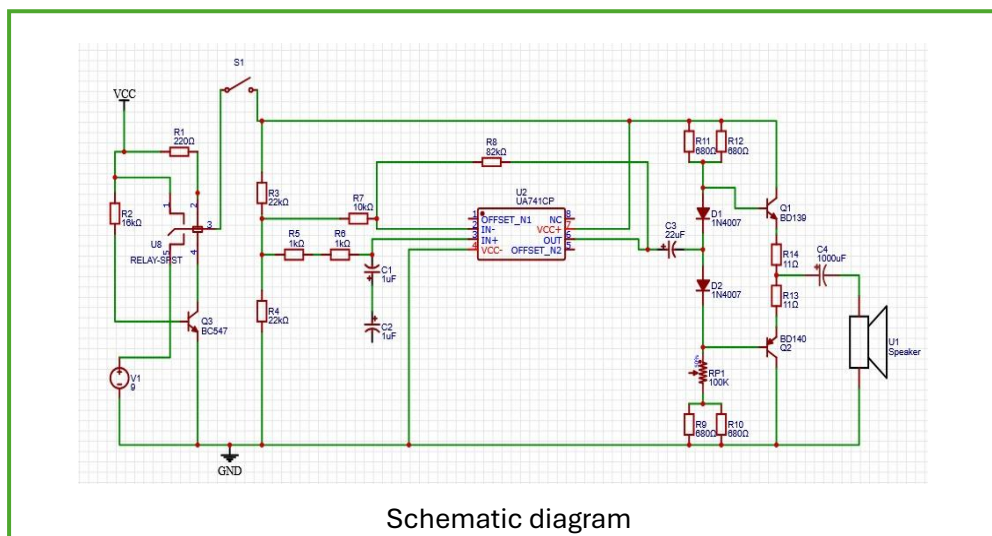
$$\text{Now the overall gain is, } \frac{1.3W}{80\mu W} = \frac{1.3}{80 \times 10^{-6}} = 16250 = 42.108(\text{in dB}).$$

At the output one 100 μF capacitor also added to attenuate the large bass frequencies.

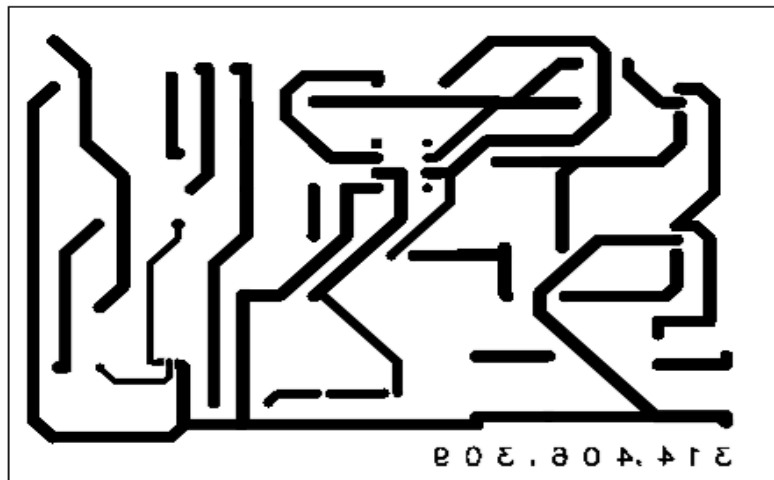
$$f_{cutoff} = 1/(2\pi * 1000 * 10^{-6} * 4) \approx 40\text{HZ}$$

6. Implementation:

A) We have designed the schematic of the circuit using 'EasyEDA' online designer.

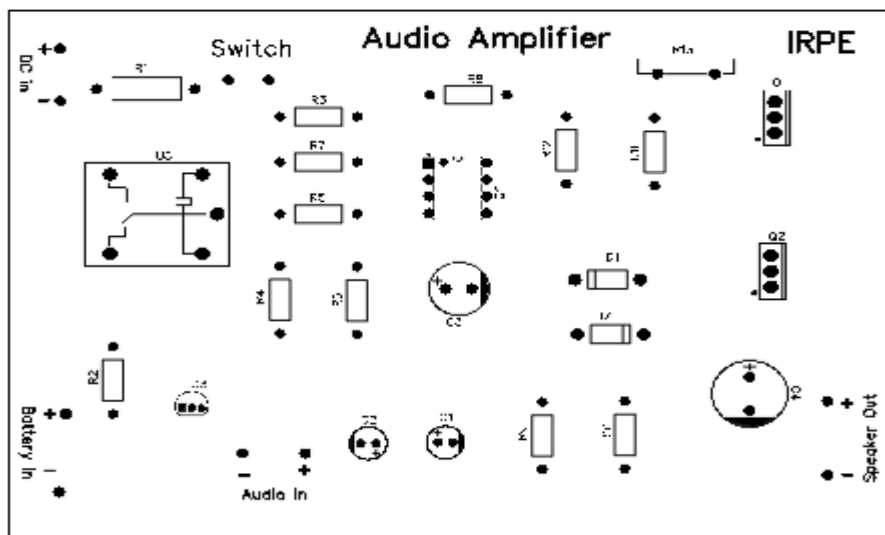


B) Then we converted our schematic design into PCB layout and then route by using the same tool.

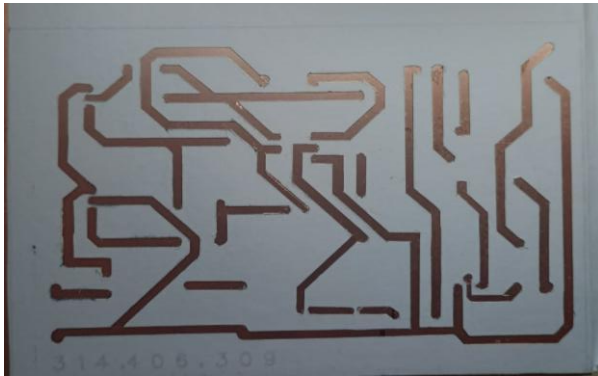


Bottom layer of PCB ↑↑

Top layer of the PCB ↓↓

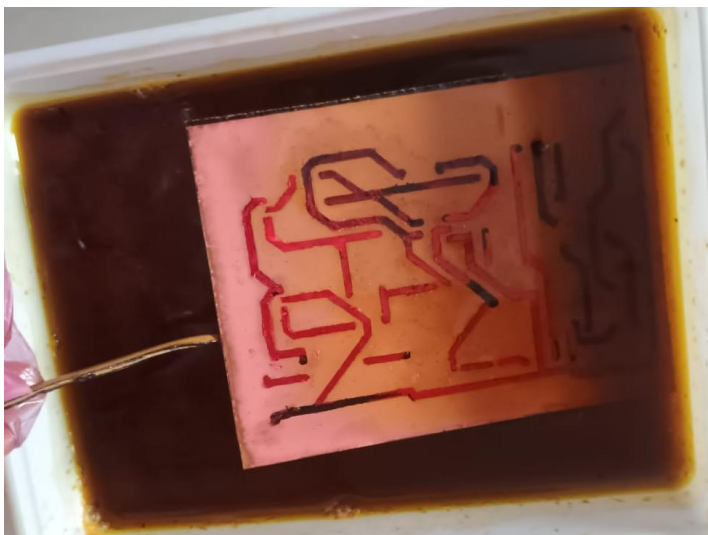


Before Etching of the PCB

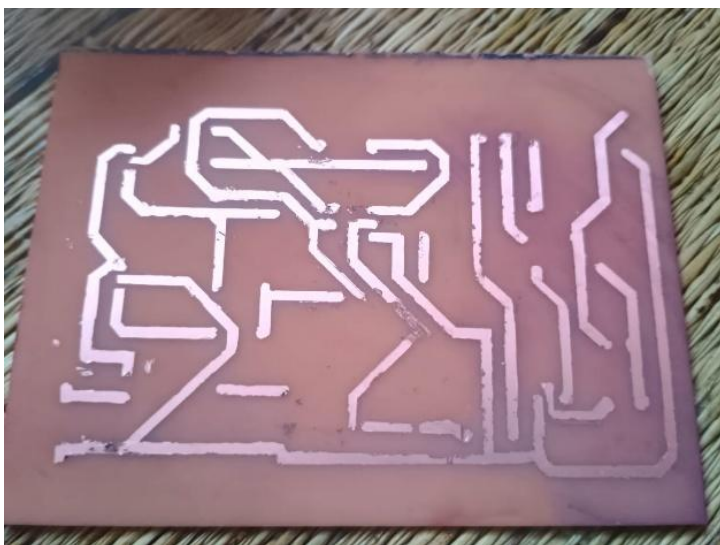


We have used FeCl_3 solution as our Etching Solution.

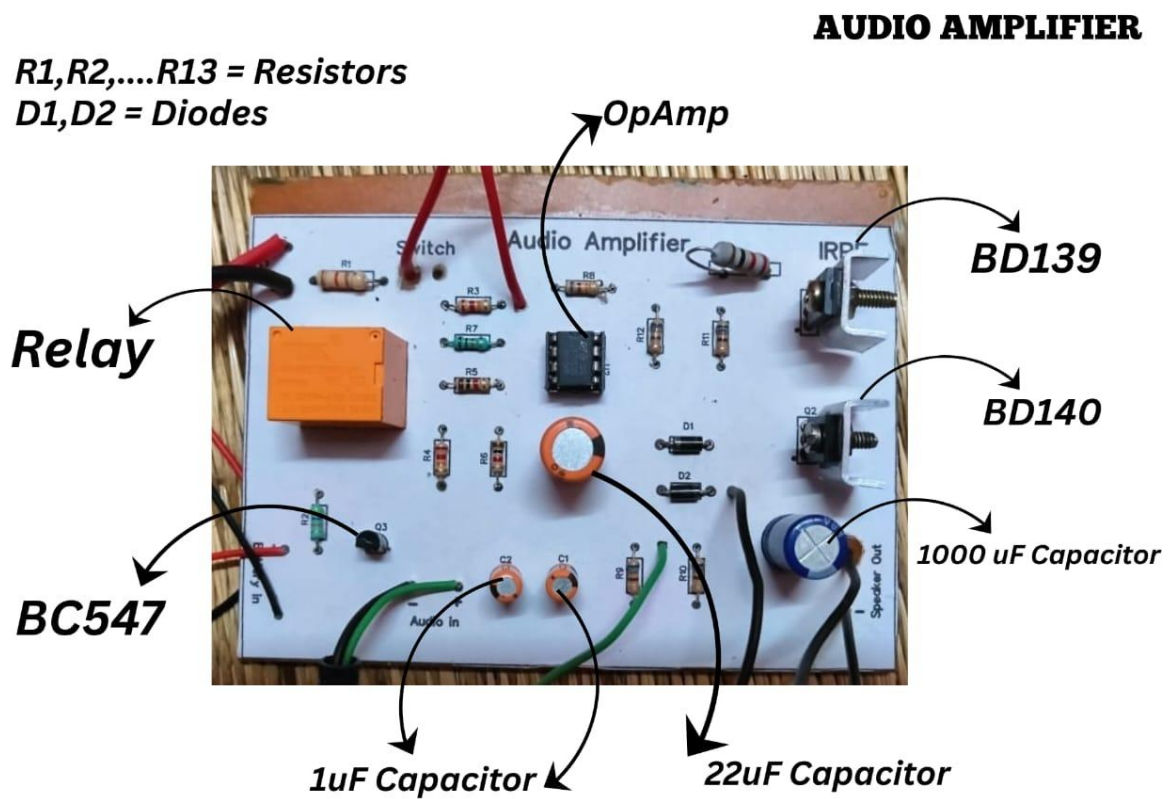
During Etching Process –



After Etching –



After Fabrication –



Final Product –



7. Equipments used –

Copper cladding board

Ferric Chloride Solution

Photo papers and markers

Soldering Iron, Wire and flux

Speaker (5W , 4 Ohm)

Resistors(22R, 120R, 680R, 1k, 16k, 22k, 10k, 82k)

Capacitors(1uF, 22uF, 1000uF)

Diodes(1N4007)

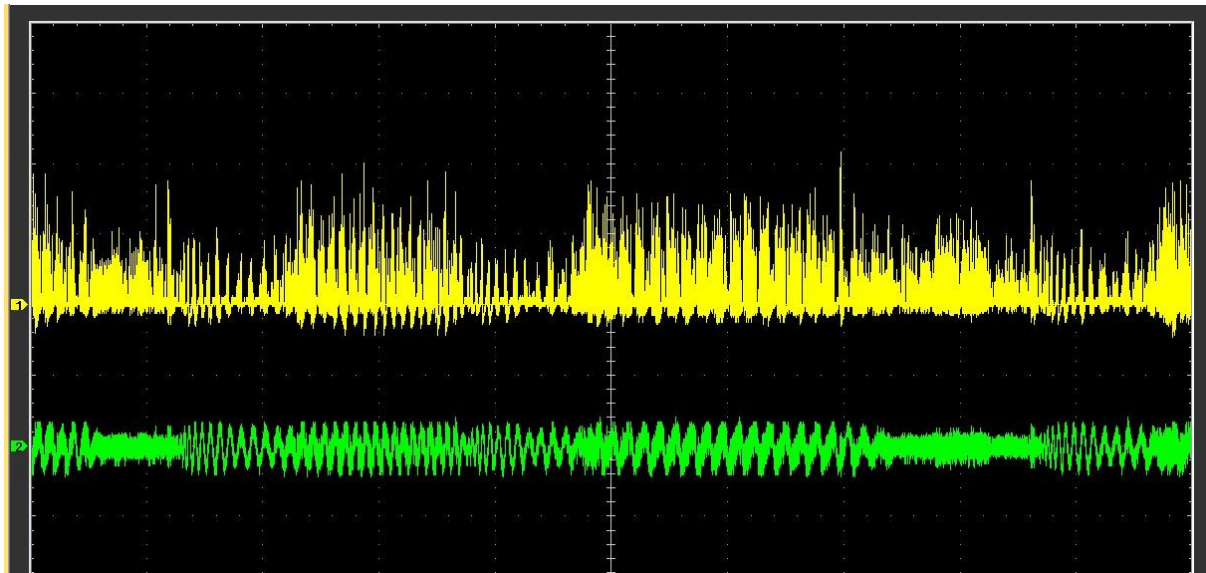
Transistors(BC547, BD139, BD140)

OpAmp(UA741)

Relay(12V) and Heat Sink

Hantek 6022BE oscilloscope

8. Result –



Input audio signal ||||| **Output Audio Signal** |||||

Power Gain of this audio amplifier is 42.108 dB.

Efficiency of this audio amplifier is around 13.3%.

This shows that input audio signal is properly amplified at the output.

9. Discussion –

The project is aimed to design, implement the Audio Amplifier. First we have used Multisim to design and simulate the circuit virtually and then we have used the EasyEDA online Designer Tool to design the PCB layout for both top and the bottom layers.

We have designed our circuit to operate in both ways using battery as well as using 12V DC from the charger output supply. So whenever we are using the direct supply from 12V, we need to isolate our circuit from the battery supply so in this case we have used a relay. To drive the relay we have used one BC547 transistor and designed the biasing in such a way so that it can operate in saturation region as a switch. Generally circuit is powered by the battery and the relay stays in normally closed position and when the supply is given from the 12V DC then the transistor turns on and drive the relay to normally open position.

For the audio voltage gain we have designed UA741 OpAmp as non inverting amplifier. To get both way swing from the (0-12)V supply we have designed a virtual ground by making a voltage divider circuit and connect OpAmp's ground connection to it.

Here we have used Electrolytic Capacitors. So we cannot directly use them in our circuit as if they are not reversed, one capacitor gets reverse voltage during AC, causing leakage, heat, oxide breakdown, and possible explosion. We reverse one electrolytic capacitor in series to handle AC signals. This prevents reverse voltage damage, making the pair act like a bipolar capacitor. Effective capacitance halves; voltage rating doubles.

We have used one 22uF capacitor in between pre amplifier and the base of the class AB power amplifier to pass all the AC signals but to prevent any DC signal to pass.

Here in our circuit we have used higher emitter resistors as only this is available to us but due to this the collector current is limited. As higher resistance is used to most of the power is dissipated across the and the efficiency drops drastically . If we use lower emitter resistances instead of 11 ohm then both the output power and the efficiency increase significantly but we are unable to find that low value resistance.

10. References:

- 1.**Electronics principles 7th edition, Albert Malvino and David J Bates.
- 2.** Microelectronic Circuit by Sedra And Smith
- 3.** Wikipedia , Google